Gemma3

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> **Demo Environment**

Development board: Jetson Orin series mainboard

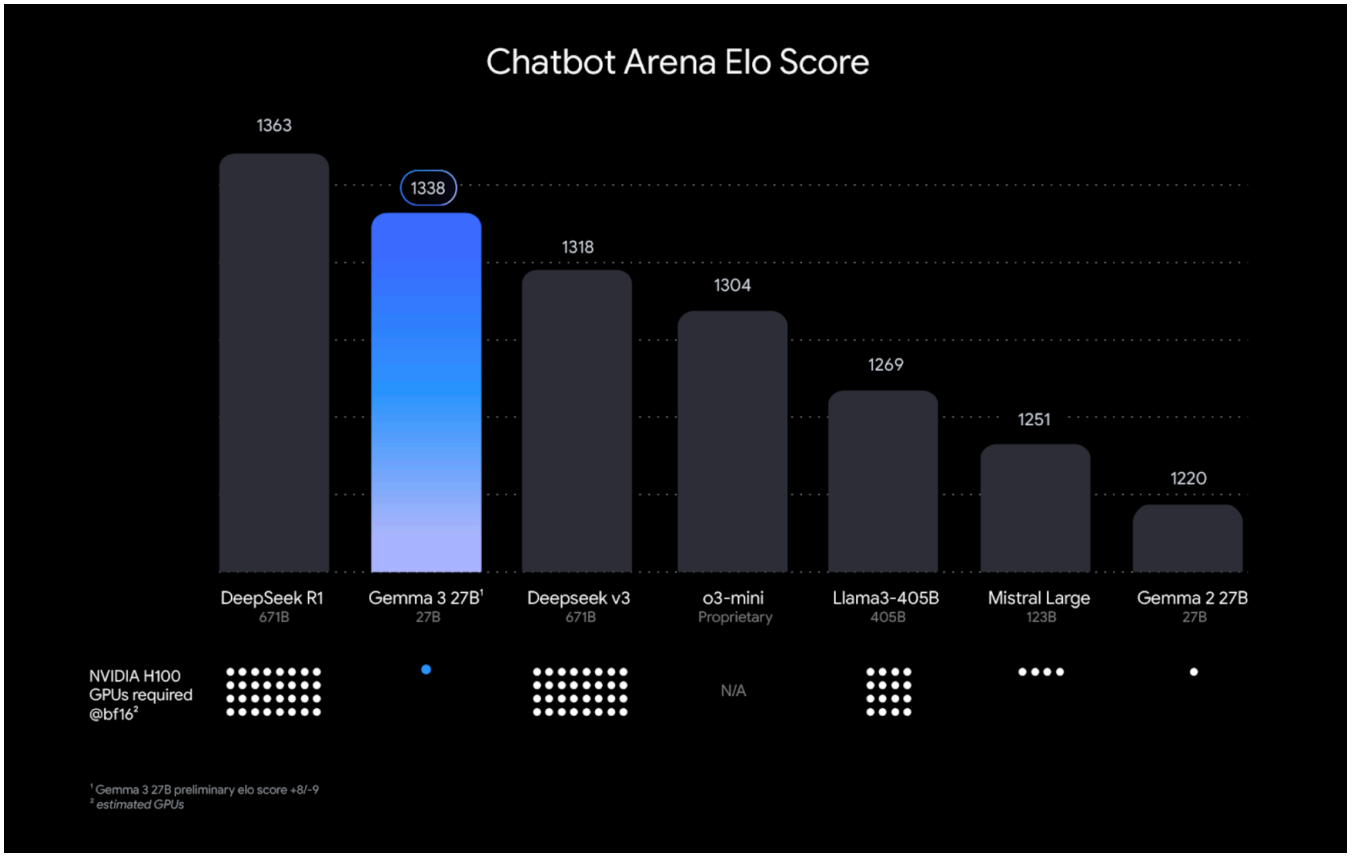
Note: Due to performance limitations, the Jetson Orin Nano 4GB needs to run the small-parameter version

Gemma is a series of lightweight models built by Google based on Gemini technology. The Gemma 3 model is a multimodal model (it can process text and images), has a 128K context window, and supports more than 140 languages.

1. Model Size

Model	Size
gemma3:1b	815MB
gemma3:4b	3.3GB

2. Performance

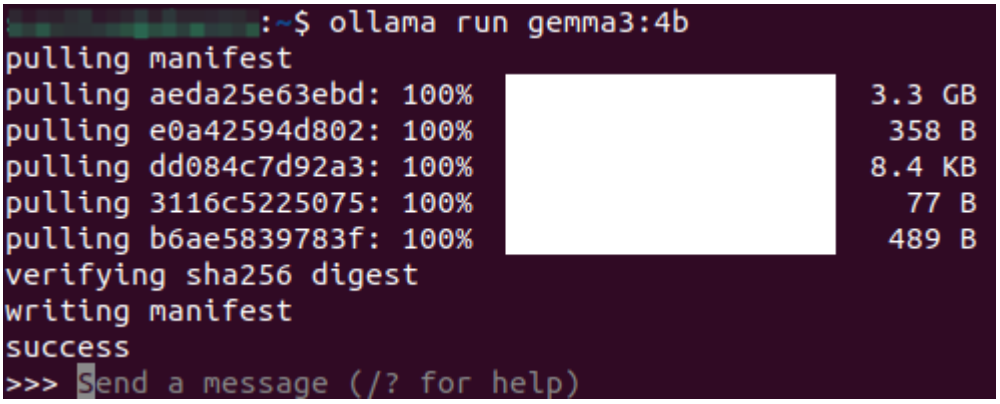


3. Using Gemma3

3.1 Running Gemma3

Run the model with the run command; if it is not downloaded yet, Ollama pulls it automatically:

```
ollama run gemma3:4b
```



3.2 Starting a Conversation

```
How to learn a programming language?
```

Response time depends on the hardware; please be patient.

```
--------
>>> How to learn a programming language?
Learning a programming language can feel daunting at first, but with a
structured approach and consistent effort, it's absolutely achievable.
Here's a breakdown of how to learn a programming language effectively,
broken down into stages and with different learning styles in mind:

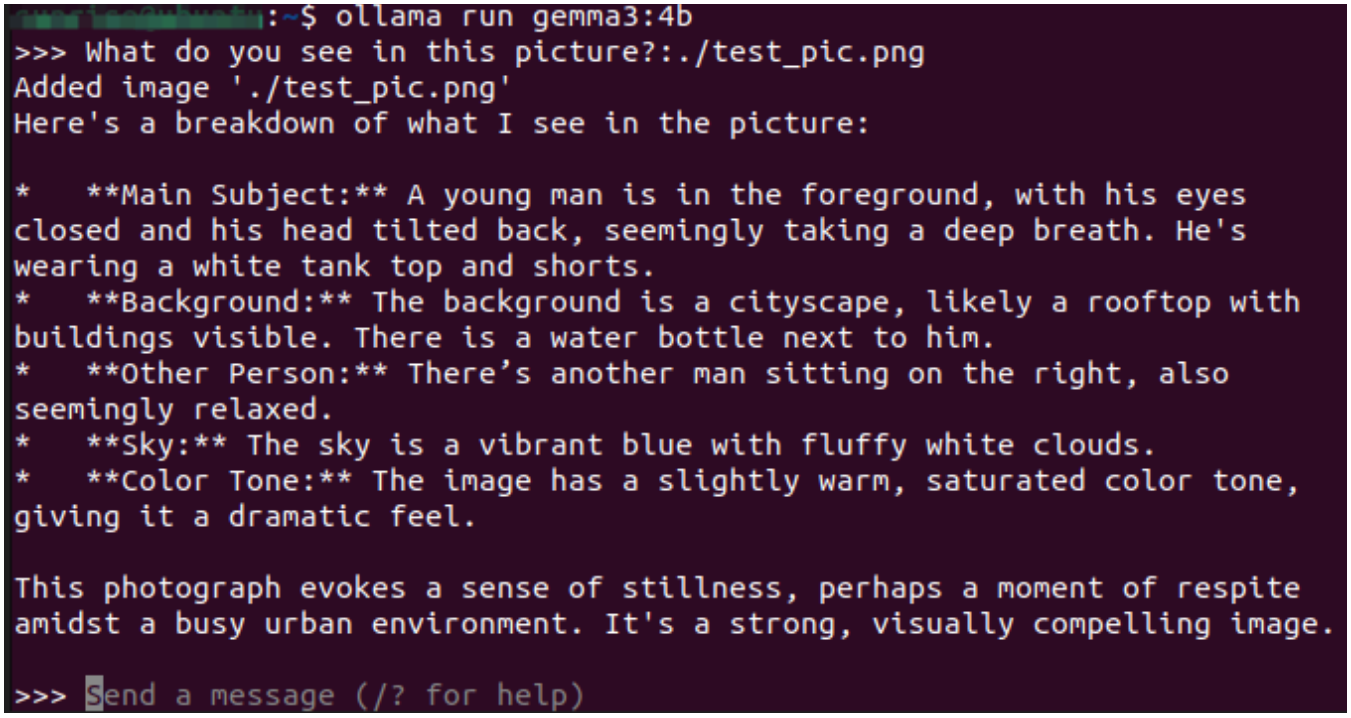
**1. Choosing Your Language:**

* **Beginner-Friendly Options:**
    * **Python:**  Highly recommended for beginners. It has a clean
syntax, huge community support, and is used in many fields (web
development, data science, scripting).
    * **JavaScript:**  Essential for web development (front-end and
increasingly back-end with Node.js).
    * **Scratch:** (Visual, block-based) A fantastic starting point for
kids and absolute beginners to understand programming concepts without
worrying about syntax.
* **Consider Your Goals:** What do you want to *do* with programming?
    * **Web Development:** JavaScript, Python (with frameworks like Django
or Flask), Ruby
    * **Data Science/Machine Learning:** Python, R
    * **Game Development:** C#, C++, Python (with Pygame)
    * **Mobile App Development:** Swift (iOS), Kotlin/Java (Android)
```

3.3 Visual Function



```
What do you see in this picture? :./test_pic.png
#Use ": + image path" in the conversation to enable the model to use its visual function and
interpret the information in the image
```



3.4 Ending the Conversation

Use the `Ctrl+d` shortcut or `/bye` to end the conversation!

References

> **Ollama**

Official website: https://ollama.com/

GitHub: https://github.com/ollama/ollama

> Gemma3

Ollama corresponding model: https://ollama.com/library/gemma3